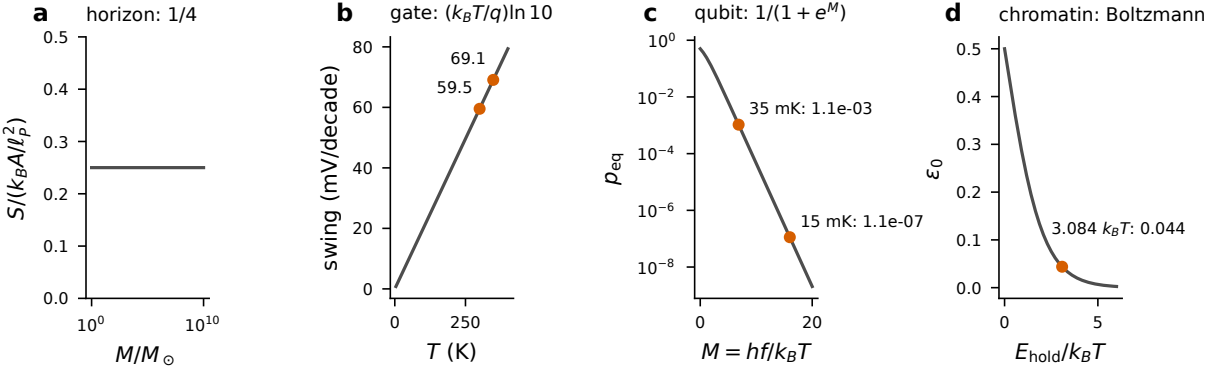
a
horizon: 1/4
$S/(k_B A/\ell_P^2)$
$M/M_\odot$
b
gate: $(k_B T/q)\ln 10$
swing (mV/decade)
69.1
59.5
T (K)
c
qubit: $1/(1+e^M)$
p_{eq}
35 mK: 1.1e-03
15 mK: 1.1e-07
$M = hf/k_B T$
d
chromatin: Boltzmann
ε_0
3.084 $k_B T$: 0.044
$E_{\mathrm{hold}}/k_B T$